

Pixel Resolution vs Intensity Resolution in Medical Imaging

Question: Differentiate how pixel resolution and intensity resolution affect image representation in a practical application like medical imaging.

Theory

Pixel Resolution: Pixel resolution refers to the number of pixels used to represent an image (spatial resolution). Higher pixel resolution preserves fine anatomical details such as blood vessels, fractures, and tissue boundaries. Lower pixel resolution reduces image sharpness and causes pixelation.

Intensity Resolution: Intensity resolution refers to the number of gray levels available to represent pixel brightness. It depends on the number of bits per pixel. Higher intensity resolution allows subtle differences in tissues to be distinguished, whereas lower intensity resolution causes loss of contrast and banding.

Practical Application in Medical Imaging

In CT, MRI, and X-ray imaging, high pixel resolution helps doctors identify small structures, while high intensity resolution helps differentiate tissues having similar densities. Both are essential for accurate diagnosis.

Comparison

- Pixel Resolution → Affects spatial detail (sharpness).
- Intensity Resolution → Affects contrast and gray-level representation.
- Low Pixel Resolution → Blurred/blocky image.
- Low Intensity Resolution → Posterization and loss of subtle tissue contrast.

Python Code

(Use the Python code provided in the question exactly as given.)

[Paste the complete code from the question here.]